

SCRIBE Data Scientist - Tech Test

Hello and welcome to the tech test for the SCRIBE Data Scientist role. Congratulations on making it this far!

The theme of this task is data extraction from herbarium sheets. A herbarium is a collection of preserved plant specimens. There are hundreds of millions of herbarium sheets in collections worldwide, which contain vital data for scientific research, including taxonomy, locality, collection events, dates, and physical specimens for morphological trait analysis.

What is the data?

We have provided a spreadsheet containing 750 herbarium sheet specimens from collections worldwide. The spreadsheet includes links to high-quality images of all 750 sheets.

For 500 of these specimens, textual data has already been automatically extracted; the remaining 250 specimens do not yet have extracted data.

The pre-extracted records can be used as reference data for exploration, comparison, supervised learning, or evaluation, but should not be assumed to be complete or error-free.

What are we asking you to do?

We would like you to develop an approach for extracting data from these herbarium sheet images. The types of data are up to you – data from the labels or the pressed specimens. We are interested in the dataset you produce, but we are just as interested in your approach: how you selected the data, what methods you used, how you assessed quality, and how you would improve the work with more time or resources. Please include some form of evaluation.

We do not expect you to process all the sheets. We expect candidates to spend no more than 2–4 hours on this task. A thoughtful, well-evaluated approach on a smaller sample is preferable to a large but poorly assessed output.

Use of tools

You may use any tools, libraries, models, or services you think are appropriate, including OCR, computer vision models, large language models, or rule-based approaches. Please describe what you used and why. If you use external AI services, including for any code-generation, please make this clear in your write-up.

What should you provide?

We will discuss your work during a 20-minute video call. You will have up to 5 minutes to present your work, followed by 15 minutes of discussion.

Please provide the following outputs:

In advance of the call – please send to arianna.salili-james@nhm.ac.uk:

- A structured dataset of your results, such as a CSV, JSON file, database export, or other format of your choosing.
- A short one-page overview, maximum 500 words, summarising:
 - what you extracted
 - the methods and tools you used
 - how you evaluated the results
 - key findings, limitations, and possible improvements
- The code you used to perform the extraction, and instructions on how we can run it.

For the call:

- A short 5-minute presentation.

How we will assess the task

We will assess your submission based on:

- the usefulness and structure of the extracted data
- the suitability of your methods
- the quality of your evaluation and critical thinking
- clarity, reproducibility, and documentation
- understanding of uncertainty, limitations, and data quality
- communication of your approach and results

We look forward to seeing what you come up with.

Good luck!

All the best,
Ben and Arianna